IT 203 – LAB 6

Nicholas Kehayias, Dustin Crawford

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Introduction:

This lab taught us how to gather information about people and systems using openly available tools and how to scan network services using Kali Linux. In Exercise 1 we used Google advanced operators, Netcraft, the Wayback Machine, the Exploit DB, CVE listings, and IP geolocation sites to see what kind of sensitive info can be found just by searching the web. Exercise 2 had us use Kali and run Nmap against a Metasploitable VM to find hosts and ports, and to try WHOIS lookups with Nmap scripts on various sites.

The goal was to learn both how attackers do reconnaissance and how defenders can spot and close those information leaks. The lab demonstrated how easy it can be to find data and interpret it online. The exercises made us more aware of the kinds of exposures organizations have and why this is crucial for their security.

Exercise 1

Task 1: Collect data on a target using Google Searches

Let's say we are an attacker who wants to find information on a new product that Google is releasing. The CEO is planning to present the product in six months. An attacker can use advanced Google Operators to gain as much information about the CEO. Open a Browser and navigate to the default Google page. See if you can answer the questions below.

Answer the following questions:

1. Who is the CEO of Google?
   Sundar Pichai
2. Where and what did he study?
   He has a Master of Science in Metallurgical Engineering from the Indian Institute of Technology, Material Science and Engineering from Stanford, and an MBA from the University of Pennsylvania.
3. What are his interests?
   Science, quizzes, reading, playing chess, sketching, and watching Bollywood movies.
4. Where does he live?
   San Francisco, California, United States
5. What car does he drive?

   Mercedes V-Class

Task 2: Find files using Google Hacking advanced operators

Head back to the main Google page. Now let's try to find something we are not supposed to have. **Just don't download it!** Let's say we want to find some free mp3s someone put on a server somewhere. We can search for site indexes by typing intitle:index of .mp3. Let's be more specific. Let's say we want to find a specific artist. You can choose your own artist. **Again, don't download**. The search should look something like this: intitle:index of Back in Black .mp3. Note that for this search, I was also able to obtain the server OS, domain, and port that it was using.

Answer the following questions:

1. What songs could you find?
   I did intitle:index of ed sheeran .mp3 and found pretty much all the songs ed Sheeran has released

2. What other data did you find?
   It also gives the date and time of each file upload/the last time it was edited.

Understanding Directory path traversals

Once you have entered a web servers file directory with a search like the one above, you can maneuver forward and backward through the directory. Click on the link at the top "parent directory" to travel one folder back in the file system. This way of traversing through the index of a site is not damaging to the server.

Index of /sounds/mp3s/Rock/ACDC

Name	Last modified	Size	Description
Parent Directory		-	
AC-DC - Problem Chil..>	2016-10-20 19:20	5.0M	
AC-DC - Who Made Who..>	2016-10-20 19:22	2.8M	
AC-DC - You Shook Me..>	2016-10-20 19:25	6.4M	
ACDC - 05 - Mistress..>	2016-10-20 19:29	5.5M	
ACDC - Back In Black..>	2016-10-20 19:30	3.9M	
ACDC - Dirty Deeds D..>	2016-10-20 19:33	3.8M	
ACDC - Got you by th..>	2016-10-20 19:35	4.1M	
ACDC - Have A Drink ..>	2016-10-20 19:38	5.5M	
ACDC - Hells Bells.mp3	2016-10-20 19:41	4.8M	
ACDC - Highway to He..>	2016-10-20 19:43	3.2M	
ACDC - Jailbreak.mp3	2016-10-20 19:47	6.4M	
ACDC - Money Talks.mp3	2016-10-20 19:50	3.4M	
ACDC - Rock 'N Roll ..>	2016-10-20 19:53	6.0M	
ACDC - Rock Your Hea..>	2016-10-20 19:56	5.6M	
ACDC - Shoot to Thri..>	2016-10-20 19:59	4.9M	
ACDC - TNT.mp3	2016-10-20 20:06	3.3M	
ACDC - The Razors Ed..>	2016-10-20 20:02	4.2M	
ACDC - Thunderstruck..>	2016-10-20 20:04	4.5M	
favicon.ico	2018-07-11 23:21	43	

Apache Server at www.sinj.com Port 443

Index of /sounds/mp3s/Rock

Name	Last modified	Size	Description
Parent Directory		-	
(6) Radiohead - High..>	2016-10-20 12:42	3.9M	
(10)Nirvana - Scentl..>	2016-10-20 12:43	3.2M	
(Soul Coughing) Supe..>	2016-10-20 12:44	3.2M	
01 - Ambulance (1).mp3	2016-10-20 12:47	5.9M	
01-the_white_stripes..>	2016-10-20 12:49	5.3M	
01_-_lazlow_-_intro-..>	2016-10-20 12:49	1.9M	
04 - young liars.mp3	2016-10-20 12:54	8.8M	
06 - Santeria.mp3	2016-10-20 12:51	3.5M	
18 Snatch Soundtrack..>	2016-10-20 12:55	4.6M	
ACDC/	2018-07-11 23:21	-	
Adema - The Way You ..>	2016-10-20 12:57	3.4M	
Adema - Trust.mp3	2016-10-20 12:58	4.0M	
Aerosmith/	2019-01-28 21:22	-	
AlbumArtSmall.jpg	2016-10-20 12:57	2.2K	
Alice In Chains - I..>	2016-10-20 12:59	3.9M	
Alice In Chains - Ma..>	2016-10-20 13:01	4.4M	
American Beauty -03-..>	2016-10-20 13:02	4.0M	
Big Wreck - That Son..>	2016-10-20 13:06	4.6M	
Blur with Radiohead ..>	2016-10-20 13:07	3.7M	
Breeders - Cannonbal..>	2016-10-20 13:08	3.3M	
Broken Social Scene ..>	2016-10-20 13:10	4.7M	
C'mon C'mon.mp3	2016-10-20 13:10	3.0M	
Cake - The Distance.mp3	2016-10-20 13:12	2.7M	
Cardigans - My Favor..>	2016-10-20 13:13	4.1M	
Chris Cornell - Can'..>	2016-10-20 13:15	3.1M	

Task 3: Filetype search query

Filetype is another advanced Google Operator. Let's say you want to find documents in pdf format. Here is the basic syntax: filetype:pdf <search words>. If an attacker wanted to find company files, they may be able to search for pdf files containing important data. For example, filetype:pdf <target> employee handbook. In this case, the attacker is looking for the company's employee handbook. There is a lot of information that can be obtained from an employee's handbook. Employee handbooks can list contact, emails, security policies, acceptable use policies, etc. Another example is filetype:pdf <target> login this can return results for default instructions on how to login to a company system.

Answer the following question:

1. Do a similar search query to the one above but this time try to find the salaries of employees in a company (Hint: Financial information are generally stored in spreadsheets)

I tried to find employe salaries for JMU, which I could not directly find, but I could find very specific teardowns of budgets and incomes for the school.

Task 4: Find data using Netcraft (https://sitereport.netcraft.com/)

Netcraft is a handy website tool that allows an attacker to gain data on a domain or web server. For example, if we type in the search box https://www.jmu.edu/ and press enter, we get network information: how long the site has been up, encryption types, hashes, server name or owner, cipher and signature algorithms, etc.

Answer the following question:

1. What is the IP of the JMU webserver and what OS is it currently running?
   The IP is 134.126.126.99 and the OS is Apache
2. What public key cipher is being used by the webserver and what is the key length?

   The public key cipher is id-ecPublicKey and the key length is 384

Task 5: Find target information using the Wayback Machine (https://web.archive.org/)

The Wayback Machine allows attackers to look at the cache of a website from previous years. Useful data can be obtained this way. Security practices were different 10, 20 plus years ago.

Complete the following task:

1. See if you can find a JMU version which has Charlie King listed as current president along with his picture on one of its pages.
   I was able to find a page with him and his picture from August of 2024
2. What was the enrollment in fall 1997?

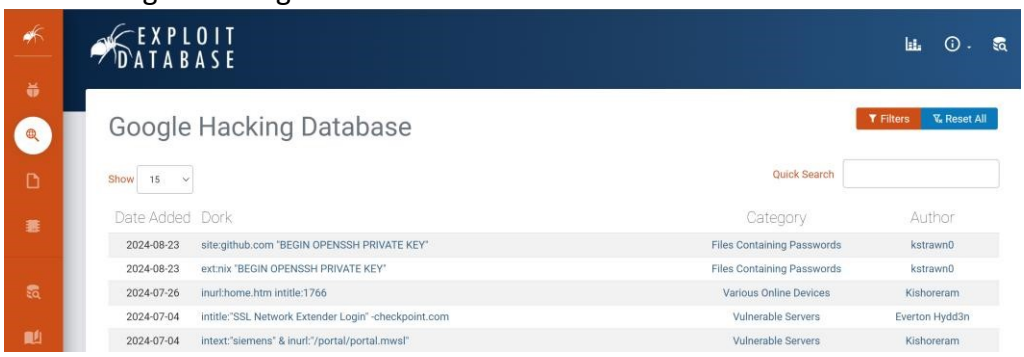
   I can't find the info on that page giving the enrollment information, but a general google search told me 11,927.

Task 6 Explore Exploit Database (https://www.exploit-db.com/google-hacking-database)

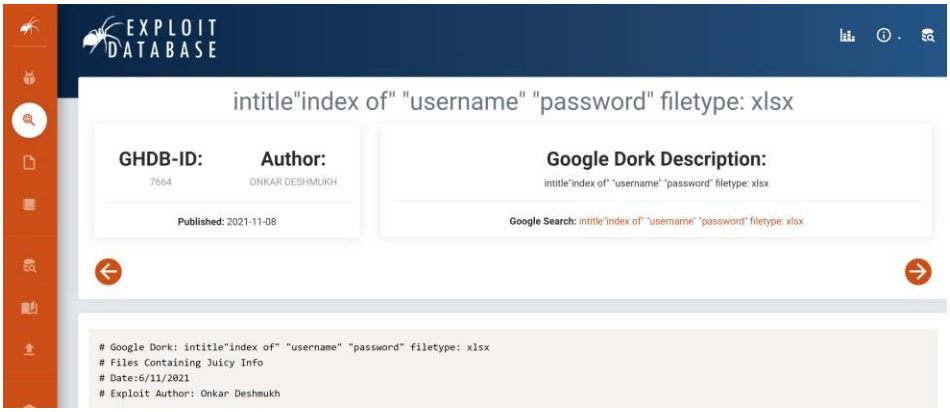
The Exploit Database lists many advanced searches that can be performed using Google. The exploits are searchable as well. Not all Google searches are. The database holds a vast number of other exploits. It also provides shellcodes that could be used to create backdoors. For this

exercise, we will be using the Google Hacking Database. Let's see what we can find. **WARNING:** Only go as far as the search. **DO NOT ATTEMPT TO LOGIN** or crack any systems found. **Do not download content**. **That would be illegal and punishable under the law.**

1. Visit the Google Hacking Database in a browser.



2. Type in the search box username.
3. Click on a link of your choice.
4. Click on the Google Search link.

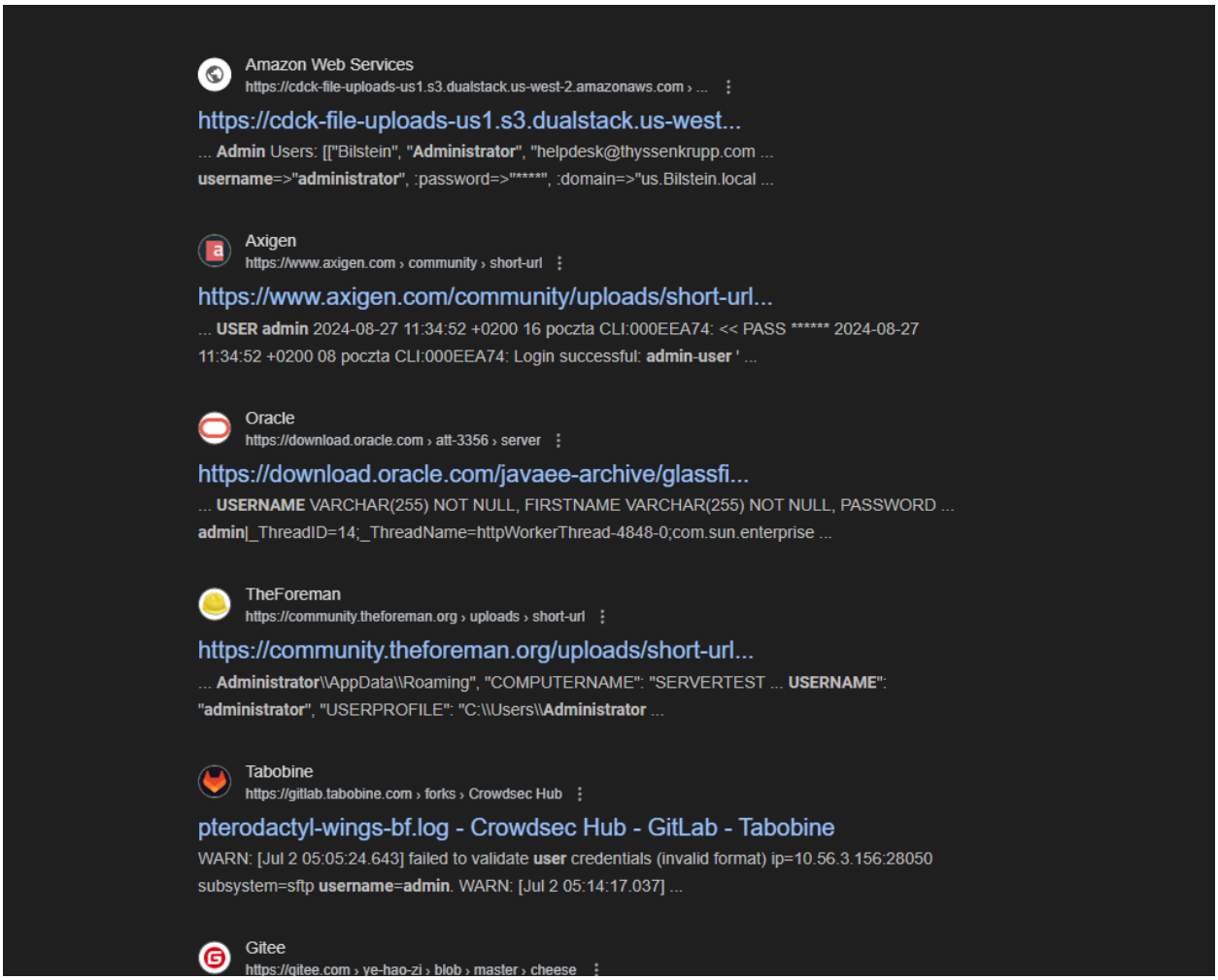


5. View the results.

There are a lot of misconfigured web servers and devices on the internet. There is also a lot of information that should be protected such as usernames and passwords. Let this be a lesson, never place unencrypted data containing usernames and passwords online.

Answer the following questions:

1. What information did you find? Make sure to take a screenshot.
   The information I found seems to be recorded logs or something of the sort for company logins, for example, there is a file listing admin users.



2. How might attackers use this information?

   They could use it to find your information, or prevent you from accessing your own information, stopping process' and creating problems.

Task 7: Explore CVE

The CVE website allows security professionals to analyze known vulnerabilities. According to the website, "CVE provides an easy-to-use web interface to CVE vulnerability data. You can browse for vendors, products and versions and view cve entries, vulnerabilities, related to them. You can view statistics about vendors, products and versions of products. CVE details are displayed in a single, easy to use page, see a sample"

Attackers can use this information as well. For example, if an attacker knows that a company is using the Linux Kernel version 2.6.32 - 4.13.1 they could search for vulnerabilities such as

CVEhttps://www.cvedetails.com/cve/CVE-2017-1000251/2017-1000251. This is a Bluetooth stack attack resulting in remote code execution in kernel space.

Answer the following questions:

1. How many vulnerabilities are in Microsoft Products in 2015?
   308 Vulnerabilities in 2015.
2. How many vulnerabilities are in Apple Products in 2015?

   351 Vulnerabilities in 2015.

Task 8: Finding the geolocation of a target

Open a browser and navigate to https://www.ipvoid.com/ip-to-google-maps/. Type in the IP address 134.126.10.50 (JMU IP address). Copy the longitude and latitude of the location. This shows the location. Take notice of the location. This is a very simple way of trying to find a targets location. If the company is using a proxy or has other security measures, then this will not be accurate. However, there are programs in Kali such as the Geolocation Tool that tend to do a better job.

Now we need to confirm that we are correct with our location. Type in the browser www.iplocation.net. In the IP Location Finder search box, type the IP of the domain you searched for above (JMU). Analyze the data and compare the two results.

The data each gave was very similar. States and cities were the same, and while the coordinates were not exactly the same, they were both extremely close that the locations are virtually indistinguishable.

Exercise 2

For this exercise you will need to run the Kali linux virtual machine and the Metasploitable. You can find them in the class folder on the network drive.

Task 1: Kali Linux on Nmap

Login to Kali Linux. Login to Metasploitable (default username/password are msfadmin/msfadmin). Find the ip address of the Metasploitable machine and record it. Open a terminal in Kali Linux and execute the following command:

```
$ nmap -T4 <ip address of metasploitable machine>
```

(Refer to the file, nmap_cheet_sheet_v7.pdf, for switch explanations.)

Answer the following questions:

1. What was the result of that operation?

```
└$ nmap -T4 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-20 16:00 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0000020s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

2. Which ports are open?

All of them are closed. They respond with a TCP RST (reset) packet.

3. What does -T4 mean?

Sets the speed template to aggressive. It speeds up the scan but might be more detectable on a network.

Task 2: Whois lookup using Nmap scripts

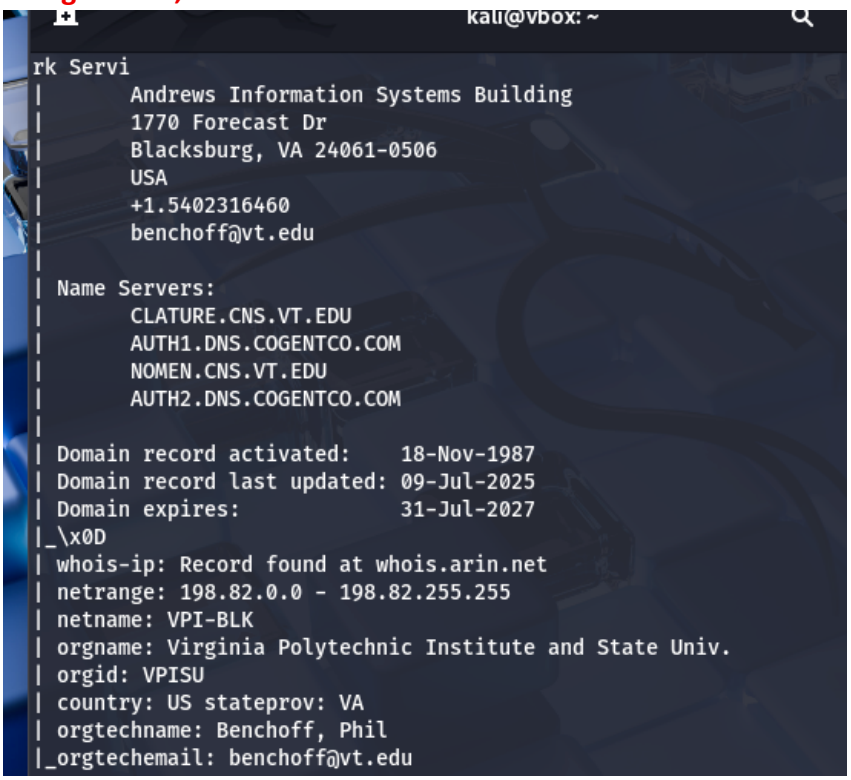
Open a terminal in Kali Linux and Type the following:

```
nmap -sn -Pn --script whois-* vt.edu
```

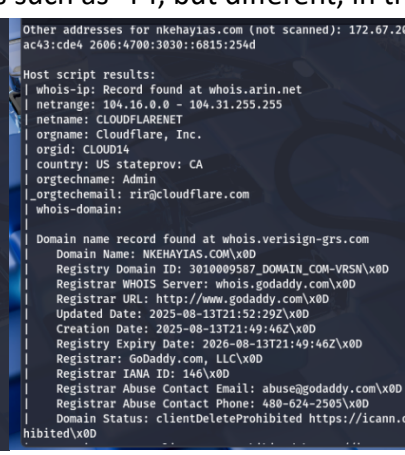
Questions:

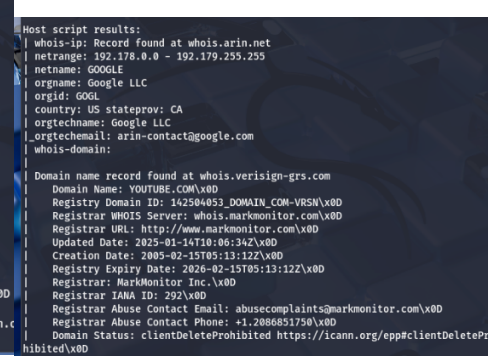
1. What information did you find?

I found domain registrant information, the technical contact, the building the web server is being hosted, and the name servers.



2. Try three more nmap commands of your choice, explain them, and report the results. (Make sure to use options such as -T4, but different, in the commands you choose.





I tested 3 other sites, one more educational one, YouTube, and my own personally hosted website. The information provided was as I had expected but very interesting on how in-depth one is able to track the information behind websites.

Conclusion

Exercise 1:

In the first section of the lab, we used Google as a basic search engine, but also using key words and public websites to access public information to demonstrate how a user might gather information to cause a cyber-attack. Basic searches like the ones done on the google CEO demonstrate how easy it is to find information. Using tools such as Netcraft, the Wayback Machine, the Exploit DB, CVE listings, and IP geolocation sites, we also demonstrated that equipped with generally very little knowledge you can gain information, data files, IP addresses, locations and more, which can be used to take advantage of people, their information, and important process'.

Exercise 2:

In the second section of the lab, we used Nmap, a powerful network scanning tool, to identify open and closed ports, detect live hosts, and gather information about domain registrants. The scan on the Metasploitable machine showed that the host was up but had no open TCP ports with all 1000 common ports closed and responding with TCP RST packets. I then performed a WHOIS lookup on vt.edu using Nmap scripts, revealing detailed domain information including the registrant organization, technical contacts, building locations, and name servers. Following this, I experimented with the nmap commands with other sites.

What to submit

Deliverables

Your report should include the following:

- A completed cover page with the report title, lab number, the name of all members of the team, Date & Signature, and an honor code statement.
- An introduction about the lab itself, explanation of what you will be doing in this lab and how you prepared for the required tasks.
- A narrative which explains the steps you are taking for each task accompanied by screenshots of the steps you are performing.
- Answers to all the questions in each section with necessary images and screenshots when applicable.
- Key learnings and takeaways from each exercise.
- A conclusion section which includes a summary of your findings, lesson learned from the lab, and your comments.
- An acknowledgement section if you have any.
- A reference section which includes all the references you have used with URL when applicable.

Grading Rubric

Grading Rubric:

Criteria	Points
Completed cover page	3
Include introduction	3
Narrative explaining the steps you are taking for each task accompanied	8
Answers to all the questions	8
Key learnings and takeaways	2
Include conclusion	3
Acknowledgement and reference	3
Total	30